

Virtual Issue Editorial

The Evolution, Status, and Research Agenda for the Future of Research in NPD Cycle Time

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Introduction

New product development (NPD) cycle time is the elapsed time from project initiation, which includes conception and definition, to the new product launch into the market (e.g., Griffin, 1997a; Kessler and Chakrabarti, 1996; Langerak, 2011). This definition, which is conceptually comparable to time-to-market, lead time, innovation time, and total time, implies that NPD cycle time can be reduced by increasing the new product's development speed, speed-to-market, or innovation speed (Cankurtaran, Langerak, and Griffin, 2013; Langerak and Hultink, 2006). We use the term NPD cycle time in this article, although the construct may be denoted differently in other articles which we cite.

Concern about NPD cycle time arose in the early 1980s when the United States was perceived as losing competitive advantage to Japanese competitors. These competitors, especially in the automotive and electronic component industries, were producing similar products of higher quality, less expensively, and moving them into the global market faster than their U.S. counterparts. The initial reaction of U.S. manufacturers was to blame their loss of competitive advantage on lower wages, and thus lower costs, in Japan. The problem was seen as a manufacturing problem, and a number of academics and consultants began to investigate how to improve U.S. manufacturing competitiveness using various new technologies and techniques, given that they could not change the difference in wage rates between the two countries.

However, early in this debate Reinertsen (1983) showed, through an analytical exercise simulating trade-offs between simultaneously achieving different

product development goals in the electronic industry, that introducing a new product six months “late” resulted in a far larger loss of profits than did commercializing a new product on time, but with higher development or manufacturing unit costs, or even with lower quality. Then in 1984 McKinsey reported that firms using CAD/CAM as a cross-functional integration system produced shorter NPD cycle times and a more significant competitive improvement than did those firms just using it as a (at the time new) manufacturing productivity tool (McKinsey and Co., 1984). These findings started to shift the focus from costs and manufacturing to product development and cycle time as the competitive problem. Indeed, several academics studying manufacturing competitiveness in the global automotive industry also concluded shortly thereafter that as much or even more competitive advantage could be obtained from speeding NPD than from just further optimizing manufacturing, publishing their work in the popular press (e.g., Takeuchi and Nonaka, 1986; Wheelwright, 1985).

The first articles on NPD cycle time published in the *Journal of Product Innovation Management* (JPIM) were by Gold (1987) and Rosenau (1988, 1989). These articles had a managerial and consulting focus and would by today's standards not be classified as research. To illustrate, Gold (1987, p. 81) “draws on his experience with product and process development in a wide array of industries to identify eight approaches to accelerating development.” Similarly, Rosenau (1988, p. 150) describes “several techniques that have not been mentioned explicitly in recent articles” that also are purported to speed NPD. Around the same time of these first JPIM articles, numerous practitioner magazines also began exhorting managers to focus on NPD cycle time to improve their firm's competitive advantage, providing examples of new products and firms that had successfully done

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so (e.g., Davis, 1989; Dumaine, 1989, 1991; Nevens, Summe, and Uttal, 1990; Sheridan, 1988; Stalk, 1988; Symonds, 1991; Uttal, 1987). In addition, a number of management books were published on the topic (e.g., Blackburn, 1991; Clark and Fujimoto, 1991; King, 1987; Rosenau, 1990; Rosenthal and March, 1991; Smith and Reinertsen, 1991).

The practitioner focus on NPD cycle time reduction also led various academics to start research

projects on the topic. In 1991 JPIM published an article by Cordero (1991, p. 283) describing “a number of time-saving techniques” and developing “a framework for managing the product supply process for speed” by making “speed a central objective of the firm, selecting faster product strategies, managing for the speedy implementation of these product strategies, and managing human resources for speed.” Again, these techniques were not derived by any apparent formal research approach, as measured against today’s standards. In 1991 JPIM also published the first formal research article with NPD cycle time as the focal dependent variable (McDonough and Barczak, 1991). This study investigated how internally developed versus externally sourced technologies and leadership style were associated with NPD cycle time. From 1992 onward JPIM began routinely publishing articles focusing on NPD cycle time and continues to do so today, although contemporary studies bear little resemblance to the earlier ones in terms of substantive content and methodological rigor.

Our article investigates the evolution and status of NPD cycle time research in terms of content, methodology, and authorships by analyzing 68 articles published in JPIM as well as four seminal articles from other journals. The insights from this analysis are used to develop an agenda for the future to ensure that the existing gaps in NPD cycle time research are recognized and tackled by future researchers. The accompanying virtual issue contains ten JPIM articles that we consider seminal on this topic, providing scholars and practitioners with a stepping stone for developing more comprehensive knowledge about the “how,” “why,” “when,” and “where” of benefiting from an accelerated NPD cycle time.

Article Selection Methodology

Creating a comprehensive list of JPIM NPD cycle time studies first involved conducting a Web of Science search using keywords including “product development,” “cycle time,” “innovation speed,” “time-to-market,” and “lead time.” The studies referencing and referenced in NPD cycle time meta-studies were also consulted (e.g., Chen, Damanpour, and Reilly, 2010; Cankurtaran et al., 2013). This effort resulted in an initial database of 117 JPIM articles. The second author developed an initial exclusion criteria protocol to screen these articles. FROM EXPERIENCE

BIOGRAPHICAL SKETCHES

Dr. Abbie Griffin holds the Royal L. Garff Endowed Chair in Marketing at the David Eccles School of Business and is the associate dean for business innovation at the School of Medicine at the University of Utah. Her research on measuring and improving the process of new product development has been published in *Marketing Science*, *Journal of Marketing Research*, *Management Science*, and *Journal of Product Innovation Management* and in the book titled *Serial Innovators: How Individuals in Large Organizations Create Breakthrough New Products*. Her 1993 article titled “Voice of the Customer” was awarded both the Frank M. Bass Dissertation Paper Award and the John D.C. Little Best Paper Award by INFORMS and has been named the seventh most important article published in Marketing Science in the last 25 years. She was the editor of the *Journal of Product Innovation Management* from 1998 to 2003. The PDMA named her as a Crawford Fellow in 2009. She was on the Board of Directors of Navistar International, a \$13 billion manufacturer of diesel engines and trucks from 1998 to 2009. Prof. Griffin is an avid quilter, hiker, and swimmer.

Dr. Fred Langerak is a professor of product development and management in the Innovation, Technology Entrepreneurship & Marketing Group of the School of Industrial Engineering at Eindhoven University of Technology in the Netherlands. He has an M.Sc. and a Ph.D. from the Erasmus School of Economics. His research focuses on managerial interventions to improve the process of conceiving, designing, developing, and bringing new products to market, and managing these products post-launch. He has published in such journals as *Journal of Product Innovation Management*, *International Journal of Research in Marketing*, *Journal Retailing*, *R&D Management*, and *Creativity and Innovation Management*.

Dr. Katrin Eling is an assistant professor of new product development in the Innovation, Technology Entrepreneurship & Marketing Group of the School of Industrial Engineering at Eindhoven University of Technology in the Netherlands. She received her Ph.D. from the same school, has an M.Sc. in Strategic Product Design from Delft University of Technology, The Netherlands, and a Diploma in Industrial Design from the University of Wuppertal, Germany. Dr. Eling’s research focuses on the successful management of the front end of new product development. Her research activities have been awarded, among others, with the Christer Karlsson Best Paper Award at the IPDM Conference 2017, the Beta Ph.D. Thesis Award 2015 by the Beta Research School for Operations Management, and Logistics and the Best Proposal Award in the 2011 Dissertation Proposal Competition of the PDMA. She has published in *Journal of Product Innovation Management*, *Creativity and Innovation Management*, and *International Journal of Market Research*.

and PERSPECTIVE articles were excluded, as they are not double blind reviewed research publications. Articles were also excluded if they focused on:

- Platform or program measures of cycle time instead of project-level measures (e.g., Chai, Wang, Song, Halman, and Brombacher, 2012; Sundgren, 1999);
- Speed of individual NPD steps, tasks, or activities, such as problem-solving speed (e.g., Atuahene-Gima and Wei, 2011), speed of flow of information (e.g., De Brentani and Reid, 2012), or transfer speed (e.g., Du, Leten, Vanhaverbeke, and Lopez-Vega, 2014); or
- Scheduling issues (e.g., Van Oorschot, Eling, and Langerak, 2018).

Other studies were excluded because their definitions of timeliness (e.g., Chryssochoidis and Wong, 2000) and schedule time adherence (e.g., Sheramata, 2002) differed distinctly from our definition of NPD cycle time or because cycle time constituted an inseparable item in a composite NPD performance measure (e.g., Song, Montoya-Weiss, and Schmidt, 1997). Finally, studies with only descriptive cycle time measures and statistics (e.g., Griffin, 1997b) were initially also excluded, but later retained (see below).

This screening resulted in a database of 71 articles, of which the second author marked five for further discussion. With the above exclusion rules in mind, the third author also screened all articles, identifying four additional ones warranting discussion. In a joint meeting, all three authors then reconsidered the exclusion criteria and together re-screened all 71 articles. Particular attention was given to the nine articles previously flagged. After a lengthy discussion, the authors decided to include in the analysis the JPIM articles that (only) report descriptive cycle time measures and statistics (e.g., Griffin, 1997b), as those numbers are indicative of the changes in NPD cycle time over time, and hence its importance to firms. These joint decisions resulted in a database of 65 articles to be included in the analysis.

A consistency check with the articles in Cankurtaran et al.'s (2013) meta-study database and articles published in JPIM referencing this meta-study identified two additional articles (Barczak, Hultink, and Sultan, 2008; Park, Lim, and Birnbaum-More, 2009) for inclusion, bringing the total to 67. As another completeness check, we looked at JPIM articles referencing the Chen et al. (2010) cycle time meta-study (which was not published in JPIM). No

additional articles were identified. During this article's writing one additional article was identified (Weiss, Hoegl, and Gibbert, 2017) bringing the total to 68 articles (see online supplement).

Coding

The initial coding scheme classified articles based on: (1) *focal grounds*, i.e., if NPD cycle time was the primary focus of the study or not; (2) *methodological grounds* distinguishing among conceptual, qualitative, and quantitative studies; (3) *variable grounds*, i.e., whether cycle time was the dependent or independent variable; and (4) *substantive grounds* in the categories of (cf. Griffin, 1997a; Kessler and Chakrabarti, 1996; Langerak, 2011): (i) theory development; (ii) tools and techniques for accelerated NPD, (iii) antecedents, (iv) risks and trade-offs in accelerated product development, (v) consequences, (vi) moderation and mediation effects, and (vii) generalizability of NPD cycle time studies. Finally, we coded the number of articles in which each author participated and drew up author networks for those with multiple JPIM articles in the NPD cycle time domain.

An article was considered focal if cycle time was the primary research focus or was a separately considered major dependent or independent variable in a larger model. Non-focal articles included cycle time more peripherally. For example, Lee and Markham (2016) included cycle time as just one of many aspects of NPD best practices investigated in the Comparative Performance Assessment Survey (CPAS). As we categorized them, qualitative research articles (e.g., Murmann, 1994) used formal qualitative methodologies (e.g., interviews, cases), while conceptual articles were more like essays (e.g., Gold, 1987), with no research method(s) presented. As we coded the quantitative articles, mostly employing the survey approach (e.g., Calantone, Randhawa, and Voorhees, 2014), we also identified a small group of articles using simulations (e.g., Van Oorschot, Langerak, and Sengupta, 2011), and created this as a new category.

In coding the articles on substantive grounds, no studies were found in the theory development category, which thus was dropped. Articles coded as tools and techniques investigate managerial actions and interventions associated with NPD cycle time (e.g., Cordero, 1991). Articles coded as antecedents

investigate the influence of factors such as project strategy, NPD process, and organizational characteristics on cycle time (e.g., Sherman, Souder, and Jenssen, 2000). Risks and trade-offs studies (e.g., Bayus, 1997) focus on interaction effects that NPD cycle time reduction might simultaneously have with other NPD objectives (e.g., development costs). The existence of trade-offs also implies that there are risks and (hidden) costs associated with accelerated NPD (Crawford, 1992). Research coded as consequences studies the influence of accelerated NPD on other NPD outcomes such as development costs, product quality, and project success (e.g., Calantone, Vickery, and Droge, 1995). A number of articles were identified that included both antecedents and consequences of cycle time reduction. Rather than classify them into both categories, they were classified into a new category: full models (e.g., Acur, Kandemir, de Weerd-Nederhof, and Song, 2010). Studies using meta-analytic procedures were classified as “generalizations” (e.g., Montoya-Weiss and Calantone, 1994). Finally, the 22 articles in the moderation/mediation category are assembled from the subgroup of the 52 articles in the antecedents, consequences, full models, and generalization categories that also examine moderating effects such as uncertainty (e.g., Bstieler, 2005) and project innovativeness (e.g., Langerak and Hultink, 2006) and/or mediating effects (e.g., Chen, Li, Evans, and Arnold, 2017) of the mechanisms through which reduced NPD cycle time and its consequences accrue.

Results

Growth of the Cycle Time Field

JPIM published the first-ever academic article on NPD cycle time in 1987. In it, Gold (1987) recounted eight approaches that his industrial experience had acquainted him with for reducing NPD cycle time. For example, contracting development externally and concurrent development, and initiating a later development stage prior to finishing the current one, were two approaches he identified. The first quantitative study on cycle time was only published four years later by McDonough and Barczak (1991). Other early articles included a conceptual article by Cordero (1991) on tools and techniques for managing the product supply

process for speed, and an impactful conceptual one by Crawford (1992), in which he argued that accelerated NPD may result in several risks and hidden costs. A firm may, for example, focus too much on developing incremental new products at the expense of radical ones that take longer to develop. An excessively tight schedule also increases the likelihood that errors are made because it may push the NPD team beyond the limits of R&D and engineering competences. There also may be additional people costs (e.g., overtime) in accelerating NPD and teams compressing NPD often consume more organizational support resources (Crawford, 1992; Langerak, 2011).

Overall, from this early start, the annual number of articles published on cycle time increased dramatically starting in 1992 (Figure 1). In the next four years (1992–1995) JPIM published a total of 19 articles. In 14 (73.7%) of these, NPD cycle time was the focal construct. The articles included conceptual and qualitative studies as well as empirical ones, and were authored by both academics and consultants. Overall, the five NPD cycle time articles of 1992 constituted 21.7% of the total articles published that year in JPIM, and in 1993 cycle time studies made up 18.5% of the total articles published, mirroring the importance the popular press was giving to accelerated NPD at the time (e.g., Dumaine, 1991; Nevens et al., 1990; Reinertsen, 1990, 1991; Smith and Reinertsen, 1991; Rosenau, 1990). Although JPIM has published at least one cycle time study in 25 of the last 32 years, Figure 1 clearly has a bimodal shape. While 37 articles were published between 1987 and 2001, none were published in 2002–2003, and then the topic has revived with 31 articles published between 2004 and 2017.

Author Networks in Cycle Time Research

The vast majority of JPIM NPD cycle time researchers have been involved in only one article. Indeed, 38 of the 68 articles published (55.9%) are one-offs by the author team. Only four author teams or networks have been associated with multiple publications: Lynn, Skov, and Abel (1999); Chen, Reilly, and Lynn (2012); and Patanakul, Chen, and Lynn (2012) and Sherman et al. (2000). Sherman, Berkowitz, and Souder (2005) and Souder, Sherman, and Davies-Cooper (1998) each have three articles in JPIM on the topic, two of which were focal and one not. Neither

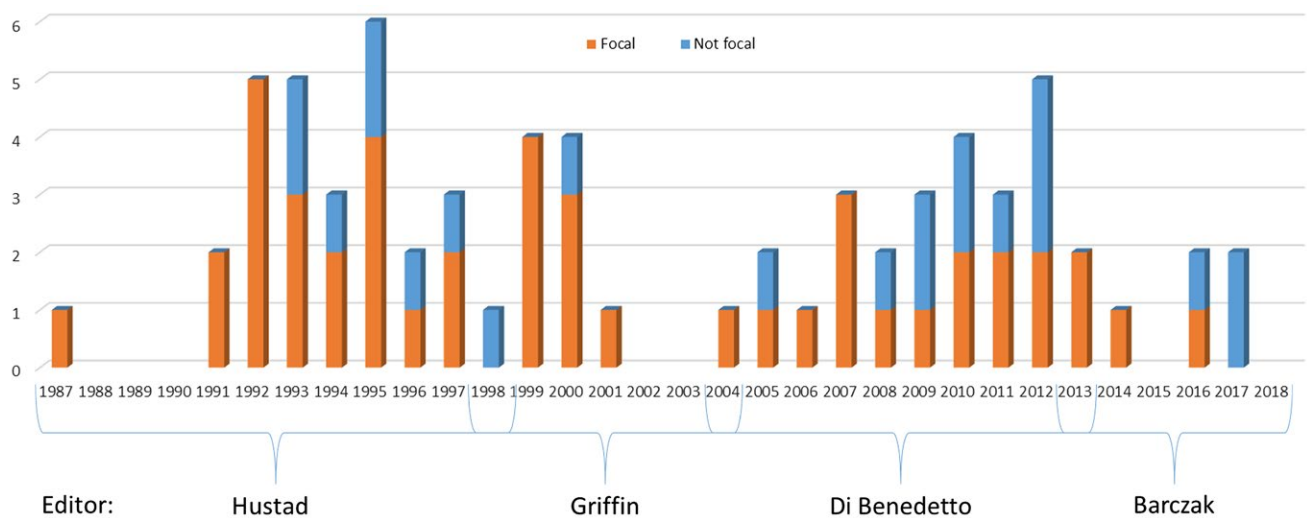


Figure 1. Focal NPD Cycle Time Articles by Year

of their networks has been active on the topic in the last five years. Both Sherman and Souder have retired and are no longer publishing.

Two JPIM NPD cycle time author networks still are active (see Figures 2 and 3). The center of the first network, Calantone, occasionally publishes research in the domain both in JPIM as well as elsewhere and, as can be seen from Figure 2, most typically with other academics in just one joint project. As his overall research interest is “product design and development processes” (<https://broad.msu.edu/facultystaff/rogercal/>, accessed November 2018), NPD cycle time is just one subdomain of many of interest to him in his overall research agenda. Vickery is predominantly a supply chain management researcher who lists one of her research interests as “time-based competition” (<https://broad.msu.edu/facultystaff/vickery/>, accessed November 2018). Like Calantone, she publishes sporadically on NPD cycle time, specifically investigating launch speed as part of a larger research agenda.

Figure 3 depicts the other still-active JPIM research network primarily focused on NPD cycle time. The McDonough/Barczak/Hultink/Griffin/Langerak group comprises 19 qualitative, empirical and (two of the three) complex simulation articles, and span all but the full model substantive content category. The network’s early core were McDonough and Barczak (1991, 1992), McDonough (1993), Griffin (1993, 1997b), Page (1993), and Griffin and Page (1993, 1996). Hultink published his first, albeit nonfocal article on cycle time in 1995 (Hultink and Robben, 1995) and through his 2006 and 2008

publications with Langerak (Langerak and Hultink, 2006; Langerak, Hultink, and Griffin, 2008) became the bridge between Barczak, Griffin, and Langerak. Langerak published his first cycle time study in 1999 (Langerak, Peelen, and Nijssen, 1999). The research activity in the network has since shifted to him and his co-authors (which still includes Griffin).

The bimodal publication pattern observed in Figure 1 may, at least partially, be explained by how authors became involved in the NPD cycle time domain and how the author networks developed over time. Of the 37 early JPIM cycle time publications, 27 (72.9%) were authored by researchers or teams with only one publication on the topic. Only McDonough, Griffin, and Page had at least three articles and only Barczak and Calantone were involved in two publications in this time period. While three-quarters of McDonough’s articles and half of Griffin’s articles were focal on the topic, none of Page’s were. McDonough was first author for both of Barczak’s publications (both focal), while Calantone only had one focal article and was the lead author for only one article.

Overall then, only McDonough and Griffin might be considered as focusing on cycle time in this early period. However, McDonough’s primary research focus has been on NPD teams and leadership. Indeed, his first three NPD cycle time articles all investigated how various team and leadership factors were associated with cycle time as the dependent variable (McDonough, 1993; McDonough and Barczak, 1991, 1992). His last publication including cycle time was in 2000, but while his earlier articles all had NPD cycle

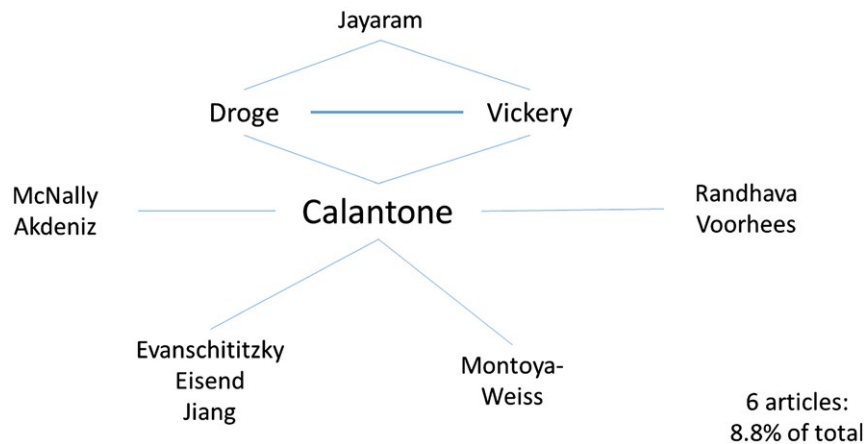


Figure 2. The Calantone Author Network

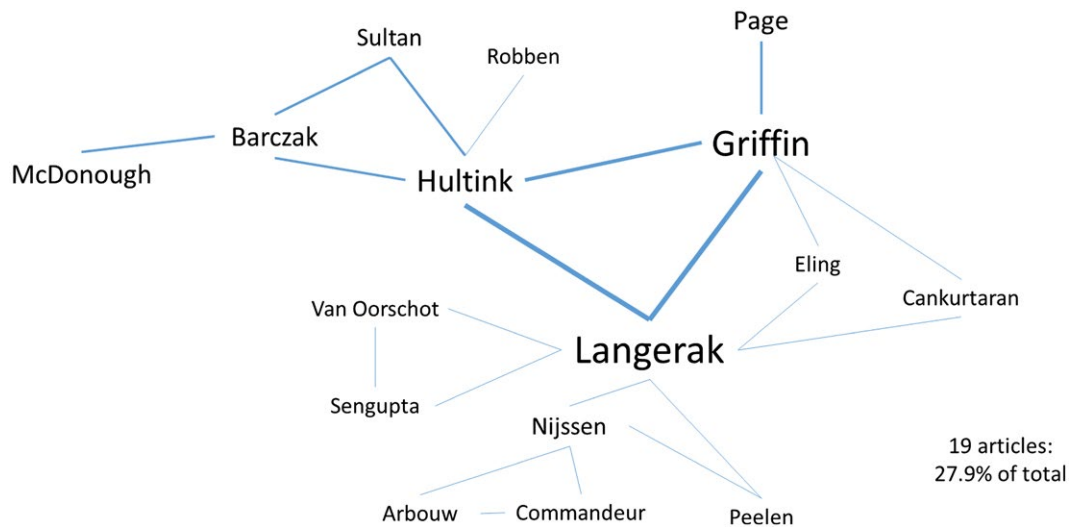


Figure 3. The Griffin, Hultink, and Langerak Author Network

time as the focal construct, this one included it only as one of several dependent variables (McDonough, 2000).

Griffin's participation in cycle time research has followed a different trajectory. She is an ongoing domain participant, with cycle time sometimes a focal construct, sometimes more peripheral. Although she has published on the topic across her academic career, she had a forced hiatus from publishing in JPIM during her editorship from 1998 to 2003. Since then, Griffin has published four additional JPIM cycle time articles, making her the most published JPIM researcher on this topic.

All of Griffin's more recent articles were co-authored with Langerak, who also has a primary interest in NPD cycle time research ([https://www.tue.](https://www.tue.nl/en/research/researchers/fred-langerak/)

[nl/en/research/researchers/fred-langerak/](https://www.tue.nl/en/research/researchers/fred-langerak/), accessed November 2018). His first JPIM publication focused on accelerated NPD (Langerak et al., 1999). Since then he has published six additional cycle time articles (16.6% of the total number of articles published since 2000). With seven articles, Langerak is the most published author in JPIM of *focal* research on NPD cycle time.

The current community of JPIM cycle time researchers thus consists of Calantone who, with various co-authors, has published five papers on the topic, mostly as part of a larger research agenda, and one domain-focused research network primarily driven by Langerak and Griffin. Altogether these two networks are responsible for 25 (36.8%) of the articles on NPD cycle time. The other 43

(63.2%) have been published by individual researchers and teams contributing an article here or there, but not focusing on it as a specific research stream, as Langerak does. Thus, cycle time clearly is not a domain that is sustained by multiple strong, active research teams.

Shifts in Focus and Content

As we have just shown, authorship has evolved and changed over the two time periods. Further analysis shows that research focus and methods used in the latter period also differ from that in the first in several ways (see Table 1).

NPD Cycle Time as the Focal versus a Non-Focal Construct. Overall, 67.6% (46) of the JPIM articles have used NPD cycle time or speed as a focal construct. However, in the early era, cycle time was focal in 75.7% (28) of the articles, while in the later time period it was the focal construct in only 58.1% (18). Thus, research has shifted from primary investigations of cycle time

to including it as one construct considered in a larger investigation of innovation. Considering that all six of Langerak's articles in the most recent time period have NPD cycle time as the focal construct, only 48.0% of the other recent articles use NPD cycle time as the focal construct.

Methods. The methods used to investigate NPD cycle time differ across the time periods. The vast majority (80.9%) of the cycle time research published in JPIM is quantitative. However, all of the three conceptual and seven qualitative research articles were published in the earlier time period. Furthermore, two of the three more complex simulation models (Langerak, Griffin, and Hultink, 2010; Van Oorschot et al., 2011) were published in the later time period. Clearly cycle time research has shifted from inductive to deductive purposes over time.

Variables. Overall, in the majority of studies (80.9%) NPD cycle time was studied as a dependent variable. However, again a shift can be observed over the two periods with 86.8% studying cycle time as a dependent

Table 1. NPD Cycle Time Article Statistics

	Total	% of total	Sum 1987–2001	Sum 2004–2017	% within time frames		% across time frames	
					1987–2001	2004–2017	1987–2001	2004–2017
# of articles	68		37	31	–	–	54.5%	45.6%
Focal construct								
Yes	46	67.6%	28	18	75.7%	58.1%	60.9%	39.1%
No	22	32.4%	9	13	24.3%	41.9%	40.9%	59.1%
Method								
Conceptual	3	4.4%	3	0	8.1%	.0%	100.0%	.0%
Qualitative	7	10.3%	7	0	18.9%	.0%	100.0%	.0%
Quantitative	55	80.9%	26	29	70.3%	93.5%	47.3%	52.7%
Simulation	3	4.4%	1	2	2.7%	6.5%	33.3%	66.7%
Cycle time variable*								
Dependent variable	55	80.9%	33	22	86.8%	59.5%	60.0%	40.0%
Independent variable	20	29.4%	5	15	13.2%	40.5%	25.0%	75.0%
Substantive content								
Tools and techniques	8	11.8%	8	0	19.0%	.0%	100.0%	.0%
Antecedents	33	48.5%	18	15	42.9%	31.3%	54.5%	45.5%
Risks and trade-offs	8	11.8%	6	2	14.3%	4.2%	75.0%	25.0%
Consequences	10	14.7%	3	7	7.1%	14.6%	30.0%	70.0%
Full models	5	7.4%	1	4	2.4%	8.3%	20.0%	80.0%
Empirical generalizations	4	5.8%	1	3	2.4%	6.2%	25.0%	75.0%
Moderation and mediation**	22	32.4%	5	17	11.9%	35.4%	22.7%	77.3%

*Some studies use cycle time as DV and IV in their models.

**Subgroup of the combined antecedents, consequences, full model, and generalization categories.

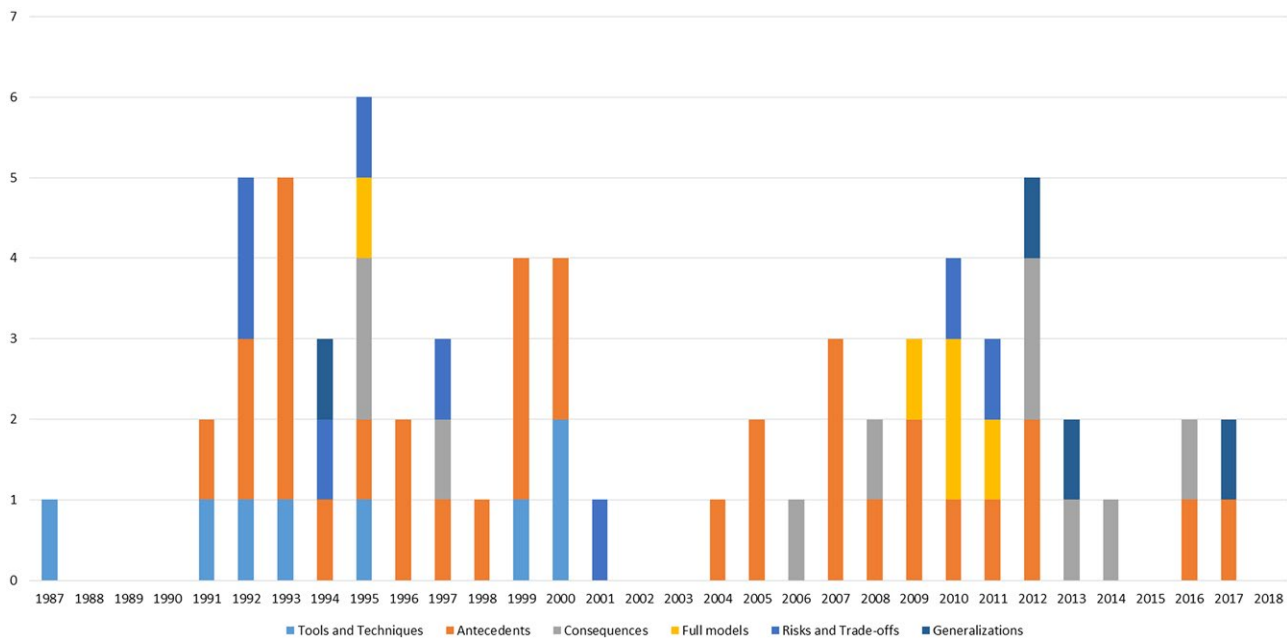


Figure 4. NPD Cycle Time Articles by Year and Content Type

variable in the early era to a more equal distribution of 59.5% (dependent) versus 40.5% (independent) in the late era. Recently, NPD cycle time as a driver of or mediator in achieving NPD success has become more important than considering accelerated cycle time as the final NPD outcome.

Substantive grounds. The article categorization by substantive category also differs between the two time frames. Table 1 shows how the substantive focus has shifted and Figure 4 graphically shows how these category types were distributed by year.

Tools and techniques. Several of the very first JPIM articles identified various tools, methods, and techniques to accelerate NPD cycle time (e.g., Cordero, 1991; Gold, 1987). In the early 1990s, Millson, Raj, and Wilemon (1992) grouped these into a hierarchy of acceleration approaches, each composed of comparable techniques to simplify development, eliminate needless NPD activities or delays, parallel NPD activities, or expedite NPD operations. Millson et al. (1992) concluded that NPD cycle time can substantially be reduced if the five approaches are thoughtfully employed in a logical order. Nijssen, Arbouw, and Commandeur (1995) delivered initial empirical support that Millson et al.'s (1992) approaches actually shorten NPD cycle time. Langerak et al. (1999) made an early inventory of 50 acceleration techniques and clustered them into nine more general NPD approaches. Using an experience survey with practitioners they conclude

(p. 174) “that managers make tradeoffs in NPD with regard to speed, development costs, and customer value” in selecting acceleration approaches, “and that not all generic NPD acceleration approaches need to be implemented in order to obtain the greatest effectiveness.”

Overall though, empirical support for NPD acceleration approach effectiveness remains limited. Moreover, and remarkably, all eight tools and techniques articles were published in the early time period. Clearly, NPD cycle time research has shifted away from investigating tools and techniques. The last study in this category, which investigated the effect of frontloading practices, was published in 2000 (Thomke and Fujimoto, 2000).

Antecedents. In parallel to identifying tools and techniques, JPIM authors started investigating NPD cycle time antecedents in the early 1990s (McDonough and Barczak, 1991). With 33 studies, investigating antecedents to NPD cycle time has constituted, and still constitutes, the majority of JPIM cycle time articles. With one study published in almost every one of the past 27 years, with the most recent published in 2017 (Chen et al., 2017), this category is also the most consistently researched in JPIM. Studies in this category test the effects of exemplar project, organizational and process characteristics on NPD cycle time. For example, Griffin (1993) and Ali, Krapfel, and Labahn (1995) show that project-level characteristics like complexity

and innovativeness lengthen NPD cycle time. Griffin (1993) even uses this finding to establish baselines to forecast project duration. Kessler and Chakrabarti (1999), in contrast, look at organizational-level cycle time antecedents, testing the effects of strategic orientation and organizational capability on NPD cycle time across industry boundaries. Sherman et al. (2000) move the topic to the process level, investigating how factors like integration of knowledge from past projects, and R&D-marketing, and R&D-customer integration reduce cycle time.

All of these studies focus on the cycle time of NPD processes in established incumbent firms. The only exception is Heirman and Clarysse (2007), who transpose these well-researched antecedent factors into the context of research-based start-ups. They find that the NPD process stage and assets at the time of founding explain differences in cycle time, but that the importance of these assets differs between software and other firms. This finding resonates with later studies conducted in established firms that find that a stage-wise approach to NPD cycle time research is warranted (Eling, Langerak, and Griffin, 2013; Stanko, Molina-Castillo, and Munuera-Aleman, 2012).

Risks and trade-offs. JPIM articles on risks and trade-offs for accelerating NPD also started early and in parallel to the tools and techniques and antecedents studies. With a total of only eight articles, it is, however, a much less popular topic. The first trade-off study (Gupta, Brockhoff, and Weisenfeld, 1992) used a conjoint analysis experiment to show that managers from different functions (e.g., R&D and marketing) make trade-offs among the criteria of NPD cycle time, development costs, and product performance. Also in 1992, Crawford published an impactful early warning of the potential hidden costs for accelerating NPD and discussed managerial actions that can “help hold down the hidden costs” (p. 197). Some years later, Bayus (1997) used a mathematical model to simulate various market, demand, and cost conditions to optimize cycle time and product performance decisions. All three studies together assume that the relationship between NPD cycle time and development costs is U-shaped, such that cycle time reduction below and above the minimum increases costs due to various factors.

The most recent trade-off studies look at managers’ use of mental models to simplify decision-making (Langerak et al., 2010; Van Oorschot et al., 2011). Such models are often applied to intervene in delayed NPD projects, because a comprehensive assessment of trade-offs across cycle time and other

metrics like costs, market entry timing, and product quality is simply not achievable. Despite these recent studies, attention appears to have drifted away from the risks and trade-offs between cycle time and other NPD goals, as only two of the eight (i.e., 25%) of these studies have been published in the second time period.

Consequences. Of these 10 articles, the majority (70.0%) have been published in the later era. Articles in this category investigate direct outcomes of NPD cycle time in terms of development cost, product quality, and overall project success. The Chen et al. (2012) generalization of past research-demonstrates that prior studies do not provide consistent evidence in favor of accelerated NPD, as also confirmed by the later Cankurtaran et al. (2013) meta-study. That’s why Chen et al. (2012) posit and then test using survey data the idea that cycle time has a curvilinear relationship with new product sales. Stanko et al. (2012), however, show that reduced cycle time linearly relates to better quality and lower costs and that, in contrast to the trade-off studies, it is not necessary to sacrifice one outcome at the expense of another. A different angle to resolving the trade-offs debate was taken by Eling et al. (2013), who use objective data to test the effects of stage-wise cycle time on new product sales. In contrast to the popular notion that just the front end stage requires more time to ensure new product quality (Crawford, 1992), they show that only consistently reducing cycle time across all NPD stages positively affects new product sales.

The consequences category is still an ongoing JPIM research topic, with the most recent (albeit nonfocal) article in 2016 (Lee and Markham, 2016) and an average of .4 studies published per year over the past 5 years (in contrast to .3 studies per year over all 32 years).

Full models. Only five articles study NPD cycle time simultaneously as a dependent (i.e., outcome) and independent (i.e., antecedent) variable. However, that 80% of these have been published in the latest time period suggests that this number might increase in the future. Notable studies include Akgun, Keskin, and Byrne (2010) and McNally, Akdeniz, and Calantone (2011). The former investigates the procedural justice climate in NPD teams, finding several antecedents with a positive impact. Procedural justice climate then positively affects team learning and development time, which in turn mediate the relationship between the procedural justice climate and new product success. McNally et al.’s (2011) study resolves contradictory findings regarding the link between cycle time and

quality, their combined effect on profitability, and their mediating roles between development cost and product profitability. They find that NPD cycle time and quality both enhance product profitability, but that the impact of cycle time is larger than that of quality. Their study further shows that cycle time and quality partially mediate the impact of front end costs on profitability, while costs incurred in later phases of the NPD process have no impact on the mediators or profitability.

Full model studies generate more in-depth understandings of the complex effects that surround NPD cycle time reduction than simpler studies of just antecedents or consequences.

Generalizations. Finally, three of the four JPIM meta-analyses have been published in the later time frame; however, cycle time is focal only in one. Cankurtaran et al. (2013) builds on prior JPIM meta-analyses (Evanschitzky, Eisend, Calantone, and Jiang, 2012; Montoya-Weiss and Calantone, 1994), as well as those in other journals (Gerwin and Barrowman, 2002; Henard and Szymanski, 2001), to update NPD cycle time consequences for six operational and external outcome dimensions of NPD success, as well as overall new product success. Their results show that NPD cycle time generally improves other NPD outcomes, but that these effects vary depending upon the cycle time measure used. Cankurtaran et al. (2013) also includes a refined consideration of Chen et al.'s (2010) antecedents. As such, JPIM's only focal NPD cycle time meta-analysis provides generalizations for both antecedents and consequences.

JPIM also has published three other meta-analyses that include cycle time as a nonfocal construct (Evanschitzky et al., 2012; Montoya-Weiss and Calantone, 1994; Weiss et al., 2017). The first two treat cycle time as one of many antecedents to NPD success. Even with data almost 20 years apart, both articles find a small-to-medium positive association between cycle time reduction and overall NPD performance. Weiss et al. (2017) focuses just on material resource adequacy as a determinant of multiple dimensions of innovation project performance, one of which is development speed, and find a positive correlation.

Since the last focal NPD cycle time generalization study was published five years ago (Cankurtaran et al., 2013), an update to these findings may be useful.

Moderation and mediation. This category consists of a regrouping of articles from the antecedents, consequences, full model, and generalization categories. As such, this category is the second biggest

of the substantive categories, with 32.4% of the 68 articles. The insights into trade-offs among NPD cycle time outcomes and inconsistencies in the results for NPD cycle time consequences may explain the recent rise (from 11.9% to 35.4%) in these moderation and mediation investigations. The articles in this category, albeit sometimes implicitly, help reconcile divergent empirical results by examining contingency effects of factors such as environmental uncertainty (Bstieler, 2005; Souder et al., 1998) and project innovativeness (Langerak and Hultink, 2006) and studying mediation effects like market entry timing proficiency (Langerak et al., 2008) and market-relating capabilities (Chen et al., 2017).

With five out of the last seven JPIM cycle time studies published containing either mediating or moderating effects, this category may drive the future of NPD cycle time research.

What Is Not Published in JPIM

Our a priori substantive domain coding scheme included theory development. However, no JPIM articles fit this category: none of the ten conceptual or qualitative JPIM articles have provided any kind of theoretical grounding for the domain. However, four articles published outside JPIM have provided important foundations for NPD cycle time theory. Emmanuelides (1993) and Zirger and Hartley (1994) published grounding work in the *Journal of Engineering and Technology Management* (JETM) and both Brown and Eisenhardt (1995) and Kessler and Chakrabarti (1996) published theory building work in the *Academy of Management Review* (AMR). Whereas, the Zirger and Hartley (1994) and Kessler and Chakrabarti (1996) articles have cycle time as the focal construct, it is an important, but not focal construct in either Emmanuelides (1993) or Brown and Eisenhardt (1995).

Emmanuelides (1993) uses a multidisciplinary lens to synthesize the literature and develop a theoretical foundation explaining the complex interactions between product development capability, NPD performance, of which cycle time is just one subdimension, and product performance. He develops 15 propositions conjecturing that innovation capabilities, both measured at the organizational and group level, impact NPD cycle time directly, and indirectly through strategic intent and contextual factors like project scope and environmental variables. Cycle time and the other NPD performance dimensions

in turn impact overall product performance. Thus, even though Emmanulides's (1993) stated purpose is developing a model of overall product development performance, his framework actually contains the first full conceptual model of antecedents, consequences, and moderating and mediating factors associated specifically with cycle time.

Zirger and Hartley (1994) extend theory by studying managerial cycle time acceleration practices. They inductively build a conceptual model linking antecedent factors of cycle time (i.e., product strategy, development process, and team structure), so-called acceleration intermediaries as mediators (i.e., project complexity, information processing capability, and motivation), and NPD cycle time. In comparison to Emmanulides (1993), their model focuses more on internal factors and excludes other contextual factors and strategic intent. Their model sets the stage for empirically testing the mechanisms through which the benefits of cycle time reduction accrue.

Brown and Eisenhardt (1995) fuse three research streams (NPD as rational plan, communication web, and disciplined problem solving) into a conceptual framework with both mediating and moderating variables intervening between the independent variables and cycle time, which in turn affects the financial success of NPD projects. Their model includes postulated effects for project leader, management, and team factors, as well as supplier involvement. They suggest that senior management, team, and supplier involvement factors have a direct effect on NPD cycle time, while project leadership has an indirect influence on cycle time through the team-level factors. In comparison to Zirger and Hartley (1994), their model is even more focused on (internal) team and process-related antecedents.

In developing their model, Kessler and Chakrabarti (1996) conducted an in-depth review of the extant cycle time literature. Their study explicitly lists different definitions of NPD cycle time and their measures, and systematically organizes and integrates past work to develop a framework linking antecedents and outcomes to cycle time reduction. In contrast to prior models, Kessler and Chakrabarti (1996) start with the external context, postulating that NPD cycle time reduction is most needed in markets with low levels of regulatory restrictiveness, and high levels of competitive

intensity and dynamism. These external forces shape the organization's strategic thrust and organizational capabilities to achieve accelerated NPD. Interestingly, they also explicate the trade-offs of reduced cycle time in terms of product quality and development costs, and their combined impact on overall project success. As such, Kessler and Chakrabarti's (1996) conceptual framework is the most comprehensive to date. Unfortunately, their set of six "propositions relating to the need for and antecedents and outcomes of innovation speed" (p. 1143) have not yet been integrally tested.

Discussion and Research Agenda

In their mapping of the overall JPIM innovation research landscape, Antons, Kleer, and Salge (2016) conclude that the topic "consequences of NPD speed" is 1 of 14 "cold" topics. Although one could challenge this conclusion because 70.0% of the studies on consequences have been published in the second time frame, of which 42.9% have been published in the past five years, this analysis shows that the NPD cycle time domain is much broader than studies focusing just on the *consequences* of NPD cycle time. The latter studies merely comprise 14.7% of the total number of cycle time studies.

Our overview of the evolution and status of cycle time research in JPIM shows that the majority (76.4%) of studies published investigate antecedents and/or consequences, while a minority (23.6%) has focused on tools and techniques, or trade-offs and risks. Interestingly, the publication pattern has a bimodal shape with two peaks in 1995 and in 2012 and a gap with no articles published between 2002 and 2003. The research content and methods used have changed substantially between the two periods, from before and after this gap. Within the first period, the focus of JPIM NPD cycle time articles was mainly on tools and techniques, antecedents and trade-offs, and NPD cycle time was mostly studied as a dependent variable (86.8%). With a stronger focus on consequences, full models, mediation and moderation effects, and with NPD cycle time more frequently being studied as an independent variable (40.5%) within the recent period, the focus of NPD cycle time research clearly has shifted from investigating substantive gaps in reducing cycle time ("how") to empirical and contextual ones in understanding the full impact of NPD cycle time reduction ("why,"

“where,” and “when”). Thus, in the past 32 years NPD cycle time has not only become established as an important outcome measure for successful product development (e.g., Millson et al., 1992) with many antecedents identified (Chen et al., 2010), but also as a key driver of NPD success with complex relationships with a number of NPD outcomes (Cankurtaran et al., 2013; Chen et al., 2012; McNally et al., 2011).

However, despite three decades of research on NPD cycle time, a holistic theory of *how*, *why*, *when*, and *where* the advantages of reduced NPD cycle time accrue is still nonexistent. Strangely, no theory development articles have been published in JPIM at all, even though JPIM is the leading academic journal devoted to NPD research, theory, and practice. Indeed, the last attempt to develop an overall theory of cycle time is over 20 years old (Kessler and Chakrabarti, 1996).

Recent empirical studies show that the relationships between antecedents, NPD cycle time and other NPD outcomes are extremely complex and depend on various project, process, organizational, and external contexts (Chen et al., 2012; McNally et al., 2011). Moreover, several new trends and perspectives have emerged (e.g., distinguishing between different types of innovation such as service or business model innovation) and new methods and tools (e.g., open and agile innovation and design thinking) have been introduced that have not yet been included in NPD cycle time theories. Thus, an up-to-date, overarching NPD cycle time theory is lacking that incorporates insights about antecedents, outcomes, and contextual conditions from early research with current NPD and innovation management theories and practices more generally. It is thus not surprising that NPD cycle time is a 2018–2020 Marketing Science Institute Research Priority: “How should firms realign for innovation and faster go-to-market?” (<http://www.msi.org/research/2018-2020-research-priorities/Organizing-for-Marketing-Agility/5.2.-external-organization/>, accessed November 2018).

We thus propose that developing and testing an updated, overarching theory of NPD cycle time is research priority number one for cycle time scholars. While, at this point, there is only one research network focusing on NPD cycle time as a focal construct and primary topic in their research, numerous individual researchers and teams occasionally investigate NPD cycle time as either a focal construct

or as part of larger overall models focused on other aspects of innovation. To encourage researchers to more routinely contribute to research on NPD cycle time and fill the gaps in understanding *how*, *why*, *when*, and *where* the benefits of NPD cycle time reduction accrue, we present a detailed future research agenda below.

An Agenda for the Future of Cycle Time Research

Table 2 presents a number of topics pertaining to each of the substantive domains or crossing them that innovation researchers may find useful to investigate.

First and foremost is the opportunity to synthesize across the meta-analytic generalizations and four grounding studies (Brown and Eisenhardt, 1995; Emmanouelides, 1993; Kessler and Chakrabarti, 1996; Zirger and Hartley, 1994) to develop an overarching theory of NPD cycle time that takes recent trends, perspectives, and developments in NPD and innovation management theory and practice into account. This theory should link antecedents and consequences of cycle time and outline the mechanisms through, and the relevant contextual conditions under which, the benefits of NPD cycle time reduction accrue. In doing so, researchers should consider differences in the level of analysis (i.e., stage, project, program, or firm) and promote consistency in operationalizing cycle time, as effects vary depending upon methodological design decisions (Cankurtaran et al., 2013).

Second, our analysis shows that the last study of acceleration tools and techniques was published in 2000 by Thomke and Fujimoto. Meanwhile, firms have adopted approaches like agile, lean, open innovation and design thinking, but how these affect NPD cycle time is as yet unknown. A recent McKinsey & Company report (Brossard, Erntell, and Hepp, 2018) also identifies new acceleration tools like digital twinning, VR-hackatons, pitch nights, and innovation garages to help incumbent firms accelerate NPD. Yet nothing is yet known about their effectiveness, either. Future research should therefore make an inventory of (new) acceleration approaches and investigate their usefulness for cycle time reduction. Research also could focus on developing additional new tools that may be required to tackle the new challenges of accelerated NPD that arise from other recent NPD trends and developments.

Table 2. NPD Cycle Time Research Agenda Suggestions*Theory Development*

- Synthesize across previous models and meta-analytic results and include latest NPD and innovation management theories and practices to develop an up-to-date and overarching theory of NPD cycle time
- Develop unified definitions and construct operationalizations of NPD cycle time
- Incorporate and link multiple levels: project, program, and firm levels of NPD cycle time

Tools & Techniques

- Identify new tools from practice: agile, lean, crowd-related techniques, open innovation, design thinking, digital twins of current products, virtual reality, hackathons, and pitch nights
- Design new tools to reduce cycle time that match current NPD and innovation management theories and practices

Antecedents

- Look across several meta-analyses on antecedents to investigate trajectories, such as relative importance changes of antecedents over time
- Identify new antecedents and update meta-analyses

Trade-offs and Risks/Consequences

- Gaining an even better understanding of the complex relations among NPD cycle time and other NPD outcomes under different contextual conditions
- Quantify how high the hidden costs of speeding actually are in different industries and whether trade-offs actually are to be made in all industries
- Investigate within and across stage trade-offs
- Multi-variate optima—Identify the global optima across dimensions of performance
- Investigate how/whether new technologies or tools, such as new manufacturing technologies, resolve trade-offs
- Investigate how trade-offs shift across the technology life cycle
- Apply more advanced modeling: agent-based, systems dynamics, other simulations

Full Models (Antecedents and Consequences)

- Extend meta-analyses to multivariate analyses in order to test full models
- Develop and test full models that also include tools and techniques and link these with resulting antecedents and with trade-offs
- Need probabilistic models

Across the Substantive Domains

- More explicitly study the *why*, *when*, *where*, and *how* of cycle time reduction for specific types of innovation, such as service innovation or business model innovation, and in different industries
- Explore how to organize for speed in open innovation contexts (when e.g., involving customers or suppliers in the process or in multi-stakeholder environments)
- Generate more insights in the role and management of NPD cycle time in start-ups in contrast to in established firms

Third, our analysis shows that the antecedents of NPD cycle time have been researched most extensively. Meta-analyses reveal numerous project strategy, NPD process, and organizational factors associated with NPD cycle time reduction. However, the results across these studies have never been systematically analyzed. Future research could therefore make a start by comparing the results of these three studies and identify and explain changes in the relative importance of particular antecedents over time, in a similar way as Evanschitsky et al. (2012) did in their meta-analysis by comparing their key success factors to those of Henard and Szymanski (2001). In addition, since the last meta-study is five years old (Cankurtaran et al., 2013), an update might be useful. Finally, as particular antecedents may be more suitable in tackling more recent challenges in reducing NPD cycle time in relation to new tools and processes such as agile, lean, or design thinking, future meta-analytic research should also be on the outlook for new generalizable antecedents.

Fourth, only a few studies have explicitly focused on risks and trade-offs in accelerated NPD. However, studies on NPD cycle time consequences also frequently discuss trade-offs. The studies in both categories provide mixed findings on the trade-offs between NPD project objectives like cost, quality, and design (cf. Gupta et al., 1992; Stanko et al., 2012), suggesting the need for a broader theory that explains the nature of the trade-offs and their impact on NPD success and also for quantifying the hidden costs of accelerated NPD. Recent advances in modelling and simulation techniques allow for more fine-grained research that simultaneously addresses multiple project outcomes as well as relevant boundary and threshold conditions. These quantifications should be conducted in different industries because trade-offs that exist in one industry may be less relevant or even nonexistent in others and optima may differ. To draw accurate conclusions, these studies need to use data from actual development teams in authentic corporate settings instead of data from contrived experimental (Gupta et al., 1992), analytical (Bayus, 1997), or simulated (Van Oorschot et al., 2011) ones. Additionally, it may also be worthwhile to explore how new technologies or tools (e.g., manufacturing processes) affect the existence of trade-offs and how trade-offs change over the life cycle of a technology, e.g., whether they diminish when the

technology matures and uncertainty is reduced. For investigating these effects, NPD cycle time needs to be studied at the program or (technology) platform level. Finally, another interesting angle for future research in the trade-off category may be to investigate how trade-offs can best be made in reducing NPD cycle time within and across different product development stages (i.e., front end, development, and launch) similar to how Van Oorschot et al. (2018) have determined gate-timing strictness and flexibility.

Fifth, developing and testing more “full” models combining NPD cycle time reduction antecedents and consequences. It may be especially interesting to directly include tool and technique usage (rather than include them indirectly through measuring the antecedents those tools are believed to affect) and also to investigate whether and, if so, which trade-offs occur depending on the antecedents and tools and techniques applied, similar to McNally et al. (2011). Finally, as previous meta-studies are based on bivariate correlations, combining meta-analysis with structural equation modeling is warranted (Cankurtaran et al., 2013).

Finally, a number of potential angles for future research should be considered that have implications across substantive domains. In the past decades, more explicit distinctions have been made in innovation management theory and practice between specific types of innovation, such as service (e.g., Storey et al., 2016), business model, and process innovation. It would thus be worthwhile to explicitly test the theories on NPD cycle time that have become established for general (product) innovation in these other contexts. Furthermore, open innovation has emerged over the past decades, in which users contribute their ideas through crowdsourcing platforms, customer or suppliers are directly involved in different NPD process stages, and/or manufacturers work collaboratively to develop new products. While open innovation has become common practice in many industries, little is known about how it impacts NPD cycle time or how companies can organize open innovation processes for speed-to-market. Last but not least, most NPD cycle time studies have been conducted in established firms, with only a few exceptions investigating cycle time in the start-up context (Heirman and Clarysse, 2007). Since start-ups and their idea-to-market time are becoming more important, especially in high-tech industries, research

is clearly needed on how to best manage start-up NPD cycle time.

Practitioners

From a practitioner perspective, this analysis of the evolution and status of NPD cycle time research, as well as our virtual issue selection of articles, is useful in explaining the virtues of accelerated NPD and the situations in which cycle time reduction is most appropriate to consider pursuing. This analysis identifies several seminal articles on tools and techniques that may facilitate making cycle time interventions (e.g., Millson et al., 1992). In addition, the meta-analytic study by Cankurtaran et al. (2013) pinpoints multiple factors at the level of the project, process, team, competencies, organization, and environment that are closely associated with cycle time reduction. This should help practitioners make appropriate interventions to successfully reduce NPD cycle time.

Virtual Issue on NPD Cycle Time

The seminal JPIM articles in the virtual issue on NPD cycle time include 10 articles that have had significant relevance for and impact in the domain and that represent the breadth of substantive, methodological, and focal grounds. Table 3 provides the positioning and details of these articles, most of which have been discussed in the preceding overview.

These articles reflect the rich variety of research on NPD cycle time published in JPIM in the past 32 years. We believe that this selection provides scholars and practitioners with a stepping stone for developing more comprehensive knowledge about the “how,” “why,” “where,” and “when” of accelerated NPD cycle time. For the theoretical grounding of NPD cycle time research, however, we refer readers to the four articles published outside of JPIM: Brown and Eisenhardt (1995), Emmanuelides (1993), Kessler and Chakrabarti (1996), Zirger and Hartley (1994). This knowledge will hopefully help scholars move the domain forward toward developing and testing an up-to-date and overarching theory on NPD cycle time, and help practitioners select the right tools and techniques to reduce NPD cycle time, while minimizing the risks, pitfalls, and trade-offs involved, as detailed in Crawford (1992) and Bayus (1997).

Table 3. Details Selected Articles for Virtual Issue on NPD Cycle Time

#	Author and Article	Substantive Domain	Methodological Approach	Substantive Focus
1.	Crawford, C. M. (1992). The hidden costs of accelerated product development. <i>JPIM</i> 9(3): 188–199.	Risks and trade-offs	Conceptual	Crawford (1992) argues that the existence of trade-offs in accelerated NPD implies five types of (hidden) risks associated with managerial practices aimed at NPD cycle time reduction.
2.	Millson, M. R., S. P. Raj, and D. Wilemon (1992). A survey of major approaches for accelerating new product development. <i>JPIM</i> 9(1): 53–69.	Tools and techniques	Qualitative	Millson et al. (1992) group NPD tools and techniques into a hierarchy of five acceleration approaches that managers can best use in a particular order to reduce NPD cycle time.
3.	Griffin, A. (1993). Metrics for measuring product development cycle time. <i>JPIM</i> 10(2): 112–125.	Antecedents	Quantitative	Griffin (1993) develops an approach for setting NPD cycle time performance baselines and shows how these baselines can be used for forecasting NPD cycle time and determining whether time reduction has been achieved.
4.	Bayus, B. L. (1997). Speed-to-market and new product performance trade-offs. <i>JPIM</i> 14(6): 485–497.	Risks and trade-offs	Simulation	Bayus (1997) uses a mathematical simulation model to optimize NPD cycle time and product performance decisions under different market, demand, and cost conditions.
5.	Kessler, E. H. and A. K. Chakrabarti. (1999). Speeding up the pace of new product development. <i>JPIM</i> 16(3): 231–247.	Antecedents (including moderation)	Quantitative	Kessler and Chakrabarti (1999) investigate the effect of antecedents factors related to the firm's strategic orientation and organizational capability on NPD cycle time.
6.	Heirman, A. and B. Clarysse. (2007). Which tangible and intangible assets matter for innovation speed in start-ups? <i>JPIM</i> 24(4): 303–315.	Antecedents	Quantitative	Heirman and Clarysse (2007) transpose well-researched antecedent factors related to cycle time reduction in the context of research based start-ups.
7.	McNally, R. C., M. B. Akdeniz, and R. J. Calantone. (2011). New product development processes and new product profitability: Exploring the mediating role of speed to market and product quality. <i>JPIM</i> 28(S1): 63–77.	Full model (including mediation)	Quantitative	McNally et al. (2011) aim to resolve contradictory findings regarding the link between NPD cycle time and product quality and their combined effect on product profitability. They also study the mediating roles of cycle time and quality in the links between development cost and cross-functional integration and product profitability.
8.	Chen, J. Y., R. R. Reilly, and G. S. Lynn. (2012). New product development speed: Too much of a good thing? <i>JPIM</i> 29(2): 288–303.	Consequences (incl. moderation)	Quantitative	Chen et al. (2012) provide an overview of past research investigating the consequences of NPD cycle time and investigate the nonlinear nature of the link between cycle time and new product sales as well as the moderating effect of uncertainty.
9.	Cankurtaran, P., F. Langerak, and A. Griffin. (2013). Consequences of new product development speed: A meta-analysis. <i>JPIM</i> 30(3): 465–486.	Generalization (incl. moderation)	Quantitative	Cankurtaran et al. (2013) conduct a meta-study to better understand the relationship between NPD cycle time and its antecedents and consequences.

(Continues)

Table 3. (Continued)

#	Author and Article	Substantive Domain	Methodological Approach	Substantive Focus
10.	Eling, K., F. Langerak, and A. Griffin. (2013) A stage-wise approach to exploring performance effects of cycle time reduction. <i>JPIM</i> 30(4): 626–641.	Consequences (including moderation)	Quantitative	Eling et al. (2013) use a stage-wise approach to NPD cycle time to test the main and interaction effects of the cycle times of the front end, development, and commercialization stages on new product performance.

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